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Microclimatic moisture and the climate change resilience of lichen epiphytes – an applied model for temperate rainforest

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ABSTRACT

The biodiversity response to climate change will, for many species, be determined by continued availability of suitable microclimatic space locally, and the ability to colonise into such refugia through dispersal. This study measured and modelled microclimatic vapour pressure deficit (VPD) for three contrasting woodland study sites, to purposefully extend microclimatic modelling to lichen epiphytes. Statistical downscaling indicated that VPD is explained by macroclimatic moisture input (precipitation) and temperature (saturated vapour pressure), with important and locally modifying effects of elevation, landscape topography, and canopy structure. Allowing for over-fitting controls, niche models for lichen epiphyte abundance were improved when including microclimatic VPD in addition to precipitation. A niche model for lichen indicators – species which determine the conservation status of UK lowland temperate rainforest – was projected at a 5 m grid-scale for a climate baseline, and three climate change scenarios (2040s, RCP 4.5; 2060s, RCP 6; 2080s, RCP 8.5), over five temperate rainforest National Nature Reserves (NNRs). Resilience of each NNR was estimated as the proportion of extant microhabitats that have an equivalent or improved microclimate (relative to the climate baseline) at incrementing spatial distances from 10 to 100 m. Despite microclimatic buffering, resilience of the NNRs was found to be limited under their current configuration. Resilience was higher for more northerly (cooler) sites, and for sites with a greater woodland extent. Resilience could be further enhanced by woodland expansion for topographically variable landscapes that have been historically deforested, with tree regeneration/planting targeted to situations that offer future microclimatic refugia.

1. Introduction

There is overwhelming scientific consensus that climate change poses a major risk to biodiversity (Arneth et al., 2020; Bellard et al., 2012; Urban, 2015). This risk has been frequently assessed using 'bioclimatic models' (Bryn et al., 2021; Heikkinen et al., 2006; Pearson and Dawson, 2003), which compare species distributions to gridded climate data for a baseline, to develop a predictive statistical model, before projecting the species response using climate change scenarios. Risk can be broadly characterised as species loss or shift in suitable climate space (Lawler et al., 2009; Markovic et al., 2014; Thuiller et al., 2005), measured against capacity for an ecological response through interrelated processes of dispersal, adaptation or acclimation (O'Connor et al., 2012). Standard gridded climate data are available for bioclimatic modelling and risk assessment at scales typically ranging from c. 1–10s km, as demonstrated by the widely used global platforms of WorldClim (Fick and Hijmans, 2017; Hijmans et al., 2005) and CHELSA (Karger et al., 2017).

There has long been awareness that many species respond directly to habitat-modified microclimates realised at small scales of metres down to millimetres (Kappen et al., 1981; Kaspari, 1993; Suggitt et al., 2011). This creates a discrepancy between both (i) climate data sourcing (standardised synoptic measurements) and (ii) the scale (c. 1–10s km) at which bioclimatic modelling has frequently operated, and the microclimates to which species respond ecologically (Kennedy, 1997; Potter et al., 2013). This discrepancy is known to have significant consequences for climate change risk assessment and conservation prioritisation across a range of taxonomic groups and ecosystem types (Maclean and Early, 2023; Nadeau et al., 2022; Suggitt et al., 2018). For example, bioclimatic modelling classically points to enhanced climate change risk for alpine species and montane ecosystems (Araújo et al., 2011; Berry et al., 2002; Thuiller et al., 2005), as temperatures get warmer, cold-adapted species literally run out of climate space at high elevations. In contrast, ecological studies have demonstrated that because of the

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complex topography in mountain environments (Scherrer and Körner, 2010, 2011; Scherrer et al., 2011), microclimatic variability realised over metres can be comparable to temperature differences calculated for macroclimatic gradients across far wider spatial and temporal scales. A downscaled approach consequently suggests that as the climate changes, a recurrent availability of locally suitable microclimates may allow species to persist in mountain environments (Randin et al., 2009). Structurally complex habitats with high levels of microclimatic variability may thus offer refugia under climate change.

As with alpine and montane ecosystems, forests and woodlands are also structurally complex, with high levels of microclimatic variability. In addition to their potential for offering refugia, recent studies have emphasised the buffering capacity of forests, for example: 1. Taking into account their landscape context (Zellweger et al., 2019), below-canopy microclimates are generally cooler in summer and warmer in winter, with lower annual temperature variation, than comparable non-canopy climates (De Frenne et al., 2019; Haesen et al., 2021), 2. Rates of below-canopy microclimatic warming are expected to be lower than for comparable non-canopy climates (De Frenne et al., 2019; De Lombaerde et al., 2022), subject to 3. Effects of stand structural dynamics, including the canopy cover affecting microclimatic outcomes and the biodiversity response to climate change (De Frenne et al., 2013; Zellweger et al., 2020). This knowledge of forest microclimates is becoming increasingly translatable into 4. Conservation-orientated management that offers microclimatic refugia, secures climate change buffering, and provides increased biodiversity resilience to climate change through adaptive management directed towards a forest's landscape context and stand structure (Frey et al., 2016; Greiser et al., 2019; Greiser et al., 2018; Sanczuk et al., 2023; Wolf et al., 2021; Zhang et al., 2021).

Given the importance of microclimates in affecting the biodiversity response to climate change (see reviews by De Frenne et al., 2021 and Kemppinen et al., 2024), this paper aims to quantify forest and woodland microclimates though with particular emphasis on the ecology of lichen epiphytes. Lichens are functionally important as epiphytes, contributing substantial canopy biomass (Berryman and McCune, 2006; McCune, 1993; Miranda-González and McCune, 2019), capturing and cycling water and nutrients (Knops et al., 1991; Van Stan and Pypker, 2015) and providing habitat and forage within food-webs (Petterson et al., 1995; Stubbs, 1989). They also contribute characteristic biodiversity to some of the planet's rarest ecosystems, including boreal and temperate rainforest (DellaSala, 2011; Ellis, 2016). However, since water availability is the principal control for patterns of dormancy and physiological activity among poikilohydric lichens (Green and Lange, 1995; Green et al., 2010; Palmqvist et al., 2010) – in contrast to previous studies that have largely focussed on forest temperature (Brusse et al., 2024; Haesen et al., 2021; Haesen et al., 2023) – the study described here focuses on forest moisture.

There is growing evidence that phenotypic differences among lichen epiphytes represent divergent adaptation, and a degree of acclimation, to patterns of moisture availability (Eriksson et al., 2018; Esseen et al., 2015; Gauslaa and Arsenault, 2020; Longinotti et al., 2017; Merinero et al., 2014), providing a functional trait-based explanation for a species' ecological niche (Gauslaa, 2014; Gauslaa and Coxson, 2011). The lichen phenotype can be expressed as the amount of precipitation (mm) per unit thallus area that is required to reach an optimum saturation for photosynthesis (Gauslaa, 2014). This supports the wide use of precipitation as a predictor in bioclimatic models for lichens (cf. Table 1 in Ellis, 2019a). However, recent research has also demonstrated a close coupling between diurnal and seasonal vapour pressure deficit (VPD) and patterns of lichen hydration and growth response (Borge and Ellis, 2024), and VPD has been proposed as a key metric in understanding the response of lichens to climate change (Stanton et al., 2022). Accordingly, this study aimed to test whether microclimatic variability in VPD might improve niche models for lichen epiphytes that are otherwise based on precipitation. Previous studies have demonstrated that microclimatic VPD can depend on landscape topography affecting moisture distribution (riparian zones, soil wetness) and forest structure affecting canopy cover (Rambo and North, 2008, 2009; von Arx et al., 2013). A nested sampling of microclimatic VPD was therefore conducted among climatically contrasting woodland study sites. Within these sites for landscape positions with different topography (slope, aspect, altitude), including differences in stand structure (canopy openness). Within these stands for contrasting microhabitats at a tree-scale (aspect and height on a tree), therefore asking:

1. How is variation in microclimatic VPD partitioned among sites, within sites among positions with different landscape topography and stand structure, and further among tree-scale microhabitats?
2. Using statistical downscaling, is microclimatic VPD predictable from macroclimate, combined with the modifying effects of landscape topography, stand structure and tree-scale microhabitats?
3. Does the addition of microclimatic VPD improve niche models for lichen epiphytes that are otherwise based on precipitation?
4. Is it possible to develop a resilience measure for lichen epiphytes, by projecting these niche models into expansive landscapes (National Nature Reserves) to estimate the gain/loss of suitable microclimates, and/or dispersal distances to local refugia under climate change?

Having explored the underpinning questions (1.-3.), the resilience measure (4.) was tested for conservation priority lichen epiphytes that characterise the UK's internationally important temperate rainforest.

2. Methods

2.1. Sampling design for microclimatic data

The sampling design aimed to measure microclimates among and within climatically contrasting study sites. All spatial analyses, here and below, were performed using ArcMap v. 10.8.2 (ESRI, Redlands, USA).

First, three climatically contrasting woodland study sites were identified by comparing the distribution of UK woodland National Nature Reserves (NNRs), with values for annual mean air temperature (T) and total precipitation (P) available as interpolated climate data at a 1 km grid-scale for the standard 30 yr reference period of 1981–2010 (Hollis et al., 2019; Perry and Hollis, 2005). The selected sites were as follows: Glasdrum NNR is located in a cooler hyper-oceanic region of western Scotland (mean temperature 7.56 °C, mean total precipitation 2737 mm); Horner Wood NNR in a warmer sub-oceanic region of southwest England (mean temperature 9.09 °C, mean total precipitation 1436 mm); Muir of Dinnet NNR in a drier and relatively more continental region of north-eastern Scotland (mean temperature 7.72 °C, mean total precipitation 755 mm) (Fig. 1). A map of hygrothermy (H) was also calculated, since this index had been used previously to measure the transition from an oceanic to more continental climate, in cross-referencing against a global definition of temperate rainforest (Ellis, 2016):

$$H = [(p_a \times t_a) / t_h - t_c] \quad (1)$$

where p_a is the mean annual precipitation (cm), t_a is the mean annual temperature (°C), and t_h and t_c are the mean temperatures of the warmest and coldest months, respectively. Hygrothermy was calculated at a 1 km grid-scale for the period of 1981–2010 (Fig. 1). The selected sites had contrasting H values of 125.1 (Glasdrum), 74.9 (Horner Wood) and 29.3 (Muir of Dinnet).

Second, four variables expected to control microclimates (Barry and Blanken, 2016; Geiger et al., 2012) were calculated from within the boundaries of each site. 1. Altitude. Sampled from a 5 m grid-scale digital terrain model (UK Ordnance Survey, Open Data Licence), 2. Heatload Index. Derived from the potential annual direct incident radiation using a function based on latitude and at a 5 m grid-scale the slope angle and aspect (McCune, 2007; McCune and Keon, 2002). Values

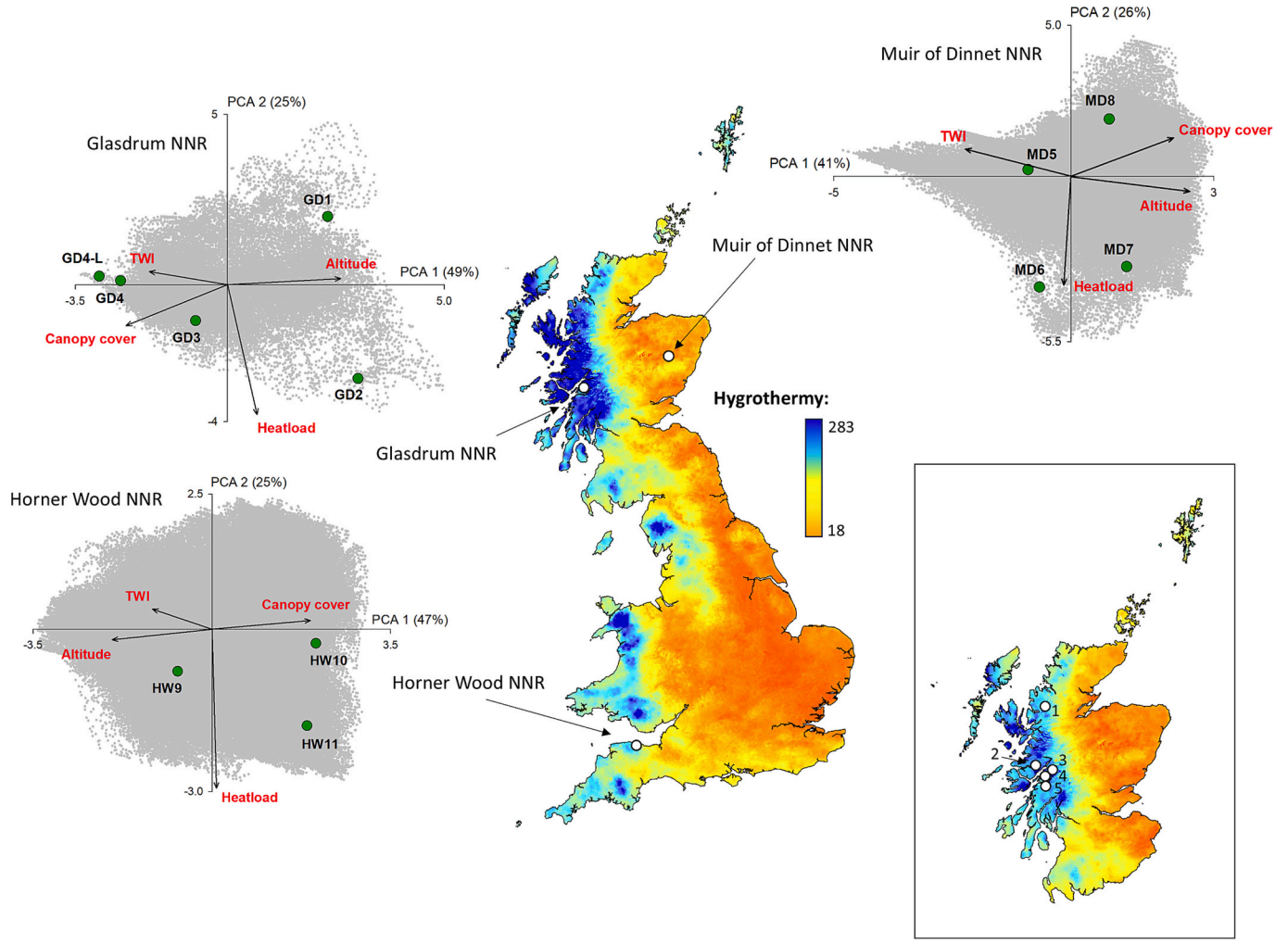


Fig. 1. Hygrothermy mapped at a 1 km grid-scale to summarise regional climatic contrasts among three woodland National Nature Reserve (NNR) study sites into which dataloggers were placed. Individual PCA plots show datalogger locations (coded green symbols) relative to the range of landscape topography and canopy cover variability within each study site (grey symbols), which are shown as vectors for altitude, heatload index, topographic wetness index, and canopy cover. Variance explained is shown for the individual PCA axes. Also showing the location of temperate rainforest NNRs for which resilience was measured under a range of climate change scenarios (inset): 1. Beinn Eighe, 2. Glencoe, 3. Ariundle, 4. Glasdrum, 5. Glen Nant. (For interpretation of the references to colour in this figure legend, the reader is referred to the online version of this article). In colour online.

for the heatload index were smoothed using a 15 m buffer (as the mean of grid cells within the buffer) to reflect the net effects of locally surrounding terrain, 3. Topographic Wetness Index. Calculated using the D8 algorithm (Beven and Kirkby, 1979) to identify at a 5 m grid-scale the steepest downslope angle from each grid cell to its neighbours, and to create a continuous flow path. Values for the topographic wetness index were smoothed using a 15 m buffer to reflect the net moisture-related effects of locally surrounding terrain, and 4. Canopy Cover. Indices of dominant leaf type (DLT) and tree cover density (TCD) were downloaded at a 10 m grid-scale for the reference year 2018 as sampled by the European Union's Copernicus Land Monitoring Service (https://land.copernicus.eu). This provides a separation between coniferous and broadleaved stands (DLT) and the proportional crown cover as vertical projection of tree crowns to a horizontal Earth surface (TCD), derived from Sentinel-1 and -2 remote sensing (Copernicus Land Monitoring Service, 2021). Focussing on broadleaved woodland the respective DLT was applied as a mask before sampling TCD. Values of TCD were smoothed using a 30 m buffer to capture an effect of edges and glades in the locally surrounding terrain, this being the scale at which lichen epiphyte distribution and community structure appear sensitive to edge effects (Boudreault et al., 2008; Esseen and Renhorn, 1998; Hilmo and Holien, 2002).

The landscape topography and canopy cover variability of each study site was summarised using ordination by principal components analysis (PCA), with the four variables (altitude, heatload index, topographic wetness index, canopy cover) centred and scaled. PCA was performed in R (R Development Core Team, 2024) using the '*prcomp*' function. This formed the basis for a selection of locations that were dispersed in ordination space, so aiming to sample a range of microclimatic contrasts within each site (Fig. 1). For selected locations, dataloggers were positioned into tree-scale microhabitats corresponding to the south-west aspects and north-east aspects of straight tree boles ($\leq$ 10° lean) at 1.3 m height above the ground, and also at 0.5 m above the ground for the south-west aspect, and in the canopy hanging from an accessible branch/twig (height and aspect being variable). For one location, dataloggers were also placed onto the upper and lower surface of a leaning tree ($\geq$ 30° lean). In each case the tree species and its diameter at breast height (DBH) was recorded, as well as percent canopy openness at the position of each datalogger using a densiometer (Englund et al., 2000; Lemmon, 1956; Paletto and Tosi, 2009). Tested among locations, there was a strong correlation between percent canopy openness measured in the field using a densiometer, and TCD derived from the Sentinel satellite imagery averaged for a 30 m buffer (Fig. S1; Pearson's product moment correlation $r = -0.922$, $P < 0.0001$ with 10

df).

In summary, four locations were selected at Glasdrum (including dataloggers positioned onto the leaning tree) and at Muir of Dinnet, and three locations at Horner Wood (Fig. 1), for a total of 46 dataloggers (Table S1). The individual dataloggers (hygrochron iButton: Maxim Integrated, California) were protected from direct wetting by being placed into a transparent and perforated hood that maintained air flow around the sensor (Fig. S2), and attached to the trees in their respective microhabitats from a horizontal nail projecting c. 2 cm from the bark surface, or when positioned in the canopy being hung from branch/twig positions with PVC-coated wire. Dataloggers recorded the temperature and relative humidity at 2 h intervals for three periods (for a total of 18 months): 1. Summer 2022 (June, July and August), 2. Winter 2022/2023 (December, January and February), and 3. An annual period (April 2023 through to March 2024). Temperature (T) and relative humidity (RH) values were used to calculate the vapour pressure deficit (VPD) for each 2 h interval (Sedano and Randerson, 2014, see also Bonan, 2008):

$$VPD = [(100 - RH)/100]^{*}[610.7^{*}10^{(7.5*T)/(237.3+T)}] \quad (2)$$

Additionally, for two locations at each study site that had a contrast in canopy cover (Table S1), extra dataloggers (Hobo MX2202: Onset Computer Corp., Massachusetts) were positioned onto the south-west and north-east aspects of their respective trees at 1.3 m height above the ground, recording the light level in Lux at 2 h intervals for the annual period (April 2023 through to March 2024). In summary, twelve dataloggers measuring light level were distributed across the study sites in locations with contrasting open and closed canopy settings. To reflect a standard unit more relevant to lichen growth, values in Lux were converted to PPFD (μmol m^{-2} s^{-1}), multiplying Lux by 0.0185 as the recommended correction factor for sunlight (Thimijan and Heins, 1983).

2.2. Microclimatic modelling

VPD was summed over consecutive 2 h intervals to a cumulative daily total for each microhabitat (VPD_{day_total}), before these daily VPD totals were further summarised as their mean per month (VPD_{month_mean}). However, before modelling VPD_{month_mean}, it was important to understand whether its derivation as a summary metric retained systematic differences in VPD measured at the high resolution spatial (microhabitat) and temporal (2 h) datalogging scales. First, with respect to VPD_{day_total}, asking whether days with contrasting values show predictable differences in their diurnal VPD. Values of VPD_{day_total} were pooled across all sites, locations, microhabitats and daily measurement periods, and percentiles calculated. Diurnal trends were plotted as the mean ± 1SE of VPD at 2 h intervals (from 00:00 to 22:00), based on the combination of days that had a VPD_{day_total} overlapping with the 99th, 95th, 90th, 75th, 50th (median), 25th, 10th, 5th and 1st percentiles. Second, with respect to VPD_{month_mean}, asking whether the days contributing to different monthly means show predictable differences in their diurnal VPD. Values of VPD_{month_mean} were pooled across all sites, locations, microhabitats, and monthly measurement periods, and percentiles calculated. Diurnal trends were plotted as the mean ± 1SE of VPD at 2 h intervals (from 00:00 to 22:00), based on the days for which their respective VPD_{month_mean} overlapped with the 99th, 95th, 90th, 75th, 50th (median), 25th, 10th, 5th and 1st percentiles.

A mixed model was used to partition variance in VPD_{month_mean} (Crawley, 2013; Zuur et al., 2009), separately for each season, at the scale of the site (reflecting regional macroclimate), the stand within a site (reflecting landscape topography and canopy structure) and the microhabitat within a stand (aspect, height, and bole lean, at the tree-scale). Mixed model analysis was performed in R (R Development Core Team, 2024) using the package '*nlme*' (Pinheiro et al., 2024; Pinheiro and Bates, 2000). The VPD_{month_mean} was also compared as a response, to nine predictor variables: 1. Mean monthly temperature, 2. Total monthly precipitation, 3. Altitude, 4. Heatload index, 5. Topographic wetness index, 6. Canopy cover, 7. Canopy phenology (see below), 8. Height above the ground, and 9. Aspect on the tree bole recorded as a categorical factor (being south-west, north-east, or upper/lower for the leaning tree).

The climate data (mean monthly temperature, total monthly precipitation) were interpolated at a 1 km grid-scale (Hollis et al., 2019; Perry and Hollis, 2005), for each respective month during which the microclimatic VPD had been recorded. Climate data were used at both the 1 km grid-scale, and as a mean to summarise across 1 km grid-squares occurring within or overlapping the boundaries of each study site. Values for total monthly precipitation were expressed as an average precipitation rate (per day), given the different number of days across months. Canopy phenology was estimated as the terrestrial chlorophyll index (TCI), originally derived from the Envisat MERIS satellite, and using woodland-specific equations that successfully predicted TCI (and canopy phenology) from average air temperatures (Boyd et al., 2011). As above, TCI values were predicted using climate data interpolated at both the 1 km grid-scale, and summarised across 1 km grid-squares occurring within or overlapping the boundaries of each study site. TCI values were plotted as a timeseries and smoothed using a running average that sampled across a window equivalent to 20 % of the data. Thresholds within the smoothed data were identified as TCI values ≥2 (closed canopy), TCI values ≤1.5 (open canopy) and intermediate values representing transitional periods of leaf-on and leaf-off (Boyd et al., 2011). Predicted TCI values were confirmed to match field observations for phenological change in canopy status (Fig. S3), with a closed canopy period being slightly shorter for the northern sites of Glasdrum and Muir of Dinnet, compared to Horner Wood, and with longer and shorter open canopy periods, respectively.

Since a non-linear response was anticipated when modelling VPD_{month_mean}, as well as a highly nested pattern of interactions among predictors, an analysis by random forest was performed in R (R Development Core Team, 2024), using the '*randomForest*' package (Liaw and Wiener, 2002). The random forest method generates an ensemble of regression trees, each of which is based on random sampling of the dataset with replacement (boot-strapping), to generate 'in-the-bag' (training) and 'out-of-the-bag' (test) data, while also using a random subset of predictors to evaluate sequential tree splitting. First, the number of trees required to minimise out-of-the-bag mean square error, and variance explained, was assessed, with the number of predictors randomly sampled for each split being set to $n = 3$ (one third of the predictor total). A run of 500 trees was sufficient to assess model error (Fig. S4A & S4B). Second, the thresholding step of the '*VSURF*' extension package (Genuer et al., 2010) was used to identify and eliminate irrelevant predictors, retaining informative predictors. Microhabitat predictors including height above the ground and aspect on the tree bole were rejected as being uninformative. Third, the optimum number of informative predictors randomly sampled for each split was tuned, by iterating from 1 to the total number of informative predictors and measuring the out-of-the-bag mean square error, and variance explained, in each case. This process was repeated 100 times to calculate a mean minimum out-of-the-bag performance corresponding to the optimised random sampling of predictors. Optimised models retained three predictors randomly sampled for each split (Fig. S4C & S4D). In a final model the out-of-the-bag mean square error was calculated, and the informative predictors were ranked in importance based on the increased mean square error when the respective predictor values were permuted (Fig. S4E & S4F). At all stages of analysis node size = 5. Prediction success was evaluated as percent variance explained, being a measure of how well the out-of-the-bag predictions (test set) explain the target variance of the training set.

The response of VPD_{month_mean} to the different predictors was examined using partial response plots, by projecting the model to the median value for all predictors, except a target predictor which was set at increments from the maximum to the minimum of observed values per season. Scenarios of open, closed and transitional canopy status were

set according to the equivalent mean temperature. For visualisation, the modelled VPD_{month_mean} was projected for each of the three study sites, according to average climate conditions experienced during the data-logging periods of summer 2022 (June, July and August) and winter 2022/2023 (December, January and February).

2.3. Microclimatic heterogeneity and resilience

A final aim was to develop a resilience score relevant to temperate rainforest lichen epiphytes at a microhabitat scale. To do this previously published inventory data (Ellis et al., 2015) was used, which had sampled epiphyte abundance at a quadrat scale (predominantly 9 × 12 cm quadrats, $n = 1013$) across 250 trees at twenty climatically contrasting sites in a clean-air region of Scotland. This same dataset has previously been compared to landscape topography and stand structural features – as proxies for the microclimate – in explaining the distribution of lichen epiphyte guilds and communities (Ellis and Eaton, 2021a, 2021b). Here, species abundance when pooled across multiple quadrats per tree, was matched to precipitation rate averaged for the winter (high moisture, low light) and summer (low moisture, high light) seasons during 2001–2010, therefore representing conditions during the decade that preceded the epiphyte inventory in 2010, using interpolated climate data at a 1 km grid-scale (Hollis et al., 2019; Perry and Hollis, 2005). This direct precipitation effect was combined with mean monthly temperature (2001–2010, 1 km grid-scale), to calculate and project VPD_{month_mean} at the microhabitat scale, the other predictors being sampled (as described above) from a 5 m grid-scale digital terrain model (UK Ordnance Survey, Open Data Licence), or using tree cover density (TCD) for the reference year 2012 (https://land.copernicus.eu).

For species which occurred on five or more trees, their abundance was compared to precipitation alone, or alternatively to precipitation plus VPD_{month_mean}. Modelling of each species' moisture response was by nonparametric multiplicative regression (McCune, 2006, 2011), implemented using Hyperniche v. 2.3 (McCune and Mefford, 2009) with a Gaussian smoother and default medium-constraints for model fitting, and with outputs assessed using cross-validated R^2 and a permutation test with 300 randomisations to calculate significance. Models for precipitation, or precipitation plus VPD_{month_mean}, were compared using cross-validated R^2. Response surfaces were plotted and visualised for a selection of species with contrasting adaptations to moisture, sense-checking model performance against prior ecological knowledge.

For the concluding analysis, abundance values were summed for lichen epiphyte species that contribute as indicators to the Lowland Rainforest Index (LRI), relevant to predominantly broadleaved woodlands, and used to evaluate Sites of Special Scientific Interest (SSSIs) in the UK conservation network (Sanderson et al., 2018). Prior to being summed, abundance values were rescaled between 0 and 1 to allow for contrasts in species thallus size, then being compared to precipitation alone, and precipitation plus VPD_{month_mean}, using nonparametric multiplicative regression (NPMR), as described above. To provide an illustrative example relevant to the UK's internationally important temperate rainforests (DellaSala, 2011; Ellis, 2016), the fitted models were projected to a downscaled surface of VPD_{month_mean} (5 m grid-scale) for exemplar NNRs within the hyper-oceanic temperate rainforest zone of western Scotland (Ellis, 2016). Three smaller and two larger NNRs were adopted (Fig. 1, inset). The smaller sites included Ariundle NNR (c. 71 ha) and Glen Nant NNR (c. 387 ha), as well as the Glasdrum NNR study site (c. 168 ha). The larger sites were Glencoe NNR (c. 5623 ha), plus the northerly and outlying Beinn Eighe NNR (c. 5857 ha). Values in VPD_{month_mean} were downscaled for each site using the random forest model (as described above), but for precipitation and temperature at a baseline (1991–2020, interpolated at a 1 km grid-scale (Hollis et al., 2019; Perry and Hollis, 2005)), and for probabilistic climate change scenarios using a 12-member ensemble (Van Vuuren and Carter, 2014; van Vuuren et al., 2011). Climate change scenarios were implemented across three degrees of severity as RCP 4.5 for the 2040s (2030–2059), RCP 6 for the 2060s (2050–2079) and RCP 8.5 for the 2080s (2070–2099), made available by the UKCP18 project as the regional average for western Scotland (Lowe et al., 2018; Murphy et al., 2018). Assuming all other features of the landscape – including woodland extent and canopy cover – remain stationary, it is possible to assess a shift in modelled abundance for LRI lichen epiphytes based on the changing microclimatic moisture regime over summer and winter seasons.

NNR resilience was scored using two measures. First, as the statistical distribution of projected abundance values for LRI lichen epiphytes at the baseline and for each climate change scenario. To acknowledge extrapolation, projections from the NPMR model were flagged where they fell below a threshold minimum neighbourhood size, set at 25 % of the model average neighbourhood size (McCune, 2006, 2011). Projections were therefore treated subjectively as having very high confidence (95–100 % of projected values meeting the threshold), high confidence (75–95 %), medium confidence (50–75 %), low confidence (25–50 %), high uncertainty (5–25 %), or extreme uncertainty (< 5 %). Second, as the minimum distance over which a given microhabitat had an equivalent or higher abundance value than its baseline, calculated for each climate change scenario using buffers of 10–100 m. This estimated the overall proportion of microhabitats for which microclimatic refugia are potentially available, at a given spatial scale.

3. Results

3.1. Sampled microclimatic data

Occasional loss of dataloggers resulted in 788 out of a total of 828 potential monthly periods with data (c. 5 % failure rate), for a maximum of 287,952 individual 2 h intervals for which VPD could be calculated. On this basis, the VPD_{day_total} calculated for a given microhabitat appeared to capture a predictable diurnal pattern of VPD (Fig. 2A). There were contrasts between diurnal VPD patterns with a high mid-day peak and relatively high night-time values (resulting in a high VPD_{day_total}), transitioning through diurnal VPD with a moderate mid-day peak with relatively low night-time values (median VPD_{day_total}), to diurnal VPD with a very weak or absence of a mid-day peak with low values throughout the day and night (low VPD_{day_total}). These diurnal VPD patterns were retained when daily totals were summarised as their VPD_{month_mean} (Fig. 2B). Microhabitats with a high VPD_{month_mean} tended to have a high mid-day peak and relatively high night-time VPD values, microhabitats with a median VPD_{month_mean} tended to have a moderate mid-day peak with relatively low night-time VPD values, and microhabitats with a low VPD_{month_mean} tended to have a very weak or absence of a mid-day peak with low VPD values throughout the day and night. In general, the variance in VPD_{month_mean} was explained in decreasing order at the scale of site > stand > microhabitat (Fig. 3), but with overlap among scales. This included periods when the variance explained by stand exceeded the variance among sites, and with the variance explained by microhabitat overlapping the variance explained by stand and being as high as 26 % of variance explained overall.

3.2. Microclimatic modelling

A random forest machine-learning approach was used to generate a predictive model for VPD_{month_mean}, with climate variables summarised within each study site, or retained as their corresponding 1 km grid-squares. Precipitation rate and mean temperature were the most important predictor variables, being modified by the topographic wetness index and woodland canopy structure (cover and phenology), with altitude and the heatload index tending to be less important (Fig. S4E & S4F). Using climate variables summarised within each study site, or retained for corresponding 1 km grid squares, the optimally tuned models had high prediction success, with the out-of-the-bag variance explained being 98 % and 97.74 %, respectively. As

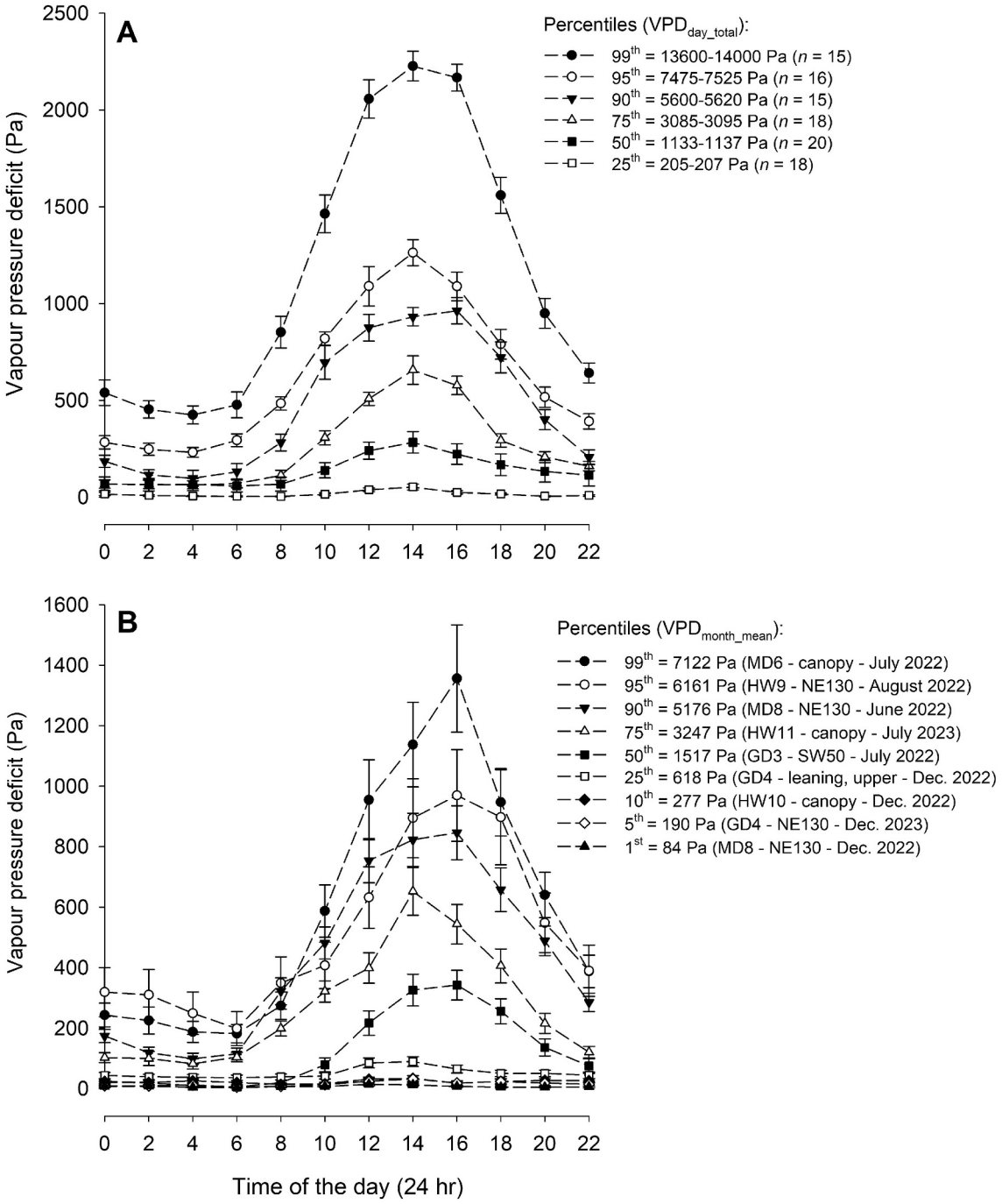


Fig. 2. Diurnal trends in VPD (mean ± 1SE) summarised for A. The combination of days that had a VPD_{day_total} overlapping with the 99th, 95th, 90th, 75th, 50th (median), 25th, 10th, 5th and 1st percentiles (range and number of days shown, noting that below the 10th percentile daily VPD total = 0), and B. Days for which their respective VPD_{month_mean} overlapped with the 99th, 95th, 90th, 75th, 50th (median), 25th, 10th, 5th and 1st percentiles (microhabitat and month shown).

demonstrated for the model using climate summarised within each study site (Fig. S5), partial response plots showed that VPD_{month_mean} had a positive relationship with mean temperature and heatload index, and a negative relationship with precipitation rate, topographic wetness index, canopy cover and altitude. To explore variability relevant to the period over which the data were sampled, the model was projected and visualised using climate summarised within each study site, for summer 2022 (June, July and August) and winter 2022/2023 (December, January and February). Patterns of VPD_{month_mean} clearly reflect climate contrasts among the study sites, and seasonal differences (different ranges), alongside the detailed effects of landscape topography and canopy structure in generating microclimatic heterogeneity (Fig. 4).

3.3. Microclimatic heterogeneity and resilience

In applying the model to characterise lichen epiphyte niches, data-loggers that measured light level demonstrated a contrast (except for the closed canopy position at Horner Wood) between lower light during what is typically the wettest period of winter conditions, and higher light during what is typically the driest period of summer conditions (Prior, 2010), confirming that in most circumstances light availability is coupled to seasonal patterns of moisture (Fig. 5). However, seasonal patterns were subject to microhabitat effects, with open canopy positions receiving more light than closed canopy positions, and south-west aspects tending to receive more light than north-east aspects of the tree

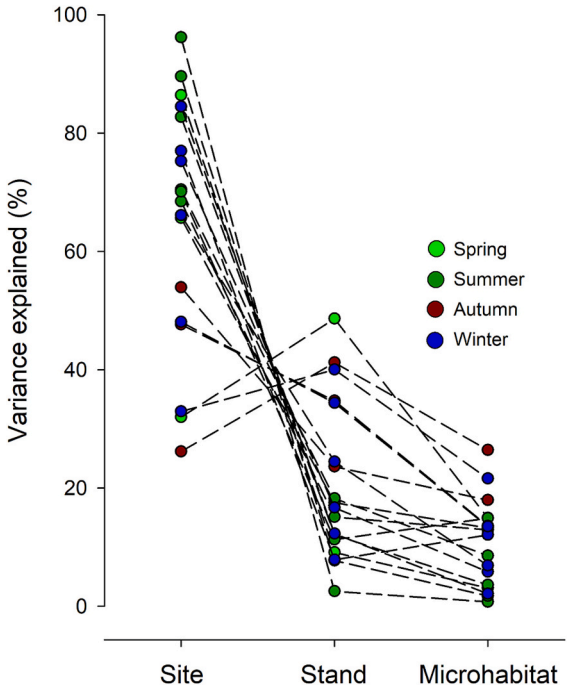


Fig. 3. Plot showing the variance in microclimatic VPD_{month_mean} explained at the scale of site, stand and microhabitat, contrasted by season. In colour online.

bole (except for the open canopy position at Muir of Dinnet).

One hundred and fifty species of lichen epiphyte were available for modelling the summer and winter moisture response (those occurring on ≥5 trees in the field sampled data). When tested using NPMR (Table S2), 101 (67 %) offered candidate models (having positive cross-validated R^2) for precipitation alone and/or precipitation plus VPD_{month_mean} as predictors, using summer season values, of which 68 % (models for 69 species) were statistically significant ($P < 0.05$). Similarly, 103 (69 %) offered candidate models using winter season values, of which 71 % (models for 73 species) were statistically significant (Table S2). The models that included microclimatic VPD_{month_mean} as a predictor tended to perform better overall than if using precipitation alone (Fig. 6). When sense-checked against prior ecological knowledge, response surfaces for precipitation and VPD_{month_mean} appeared ecologically robust (Fig. 7), for example, with contrasts among species consistent with their phenotypic adaptation (see *4.2 Microclimatic Modelling of VPD and the Epiphyte Response*, below), exemplified for: 1. *Bryoria fuscescens*, 2. *Lobaria pulmonaria*, and 3. *Nephroma parile*.

NPMR was applied to an assemblage of species contributing to the UK's SSSI Lowland Rainforest Index (LRI), with sufficient data being available for a subset of nine out of a potential 56 lichen epiphyte indicators: *Arthonia ilicina*, *Arthonia vinosa*, *Hypotrachyna taylorensis*, *Lobarina scrobiculata*, *Lopadium disciforme*, *Megalospora tuberculosa*, *Micarea stipitata*, *Nephroma parile*, and *Pyrenula occidentalis*. Models that included microclimatic VPD_{month_mean} as a predictor again performed better than using precipitation alone – cross-validated R^2 was 0.051 compared to 0.046 (summer season), and 0.060 compared to 0.056 (winter season), respectively – being statistically significant for both the summer and winter seasons ($P = 0.007$ in both cases). Response surfaces for precipitation and VPD_{month_mean} (Fig. 7) demonstrated a positive response to higher precipitation and lower VPD_{month_mean} for the summer season. For the winter season there was a unimodal response which peaked for medium-high values of precipitation with low VPD_{month_mean}, then declined towards the very wettest habitats combining the highest precipitation with the lowest VPD_{month_mean}. There was a weak bimodality with higher values for medium-low precipitation values and higher VPD_{month_mean} which could reflect a bias in the subset of species available for modelling.

Projecting these models for five temperate rainforest NNRs under baseline and climate change scenarios (Fig. 8) showed that: 1. Confidence in projected abundance values for LRI indicators tended to decline as climate change scenarios become more extreme, representing extrapolation into climate space outside of the training data. Generally, there was very high confidence for the 2040s under RCP 4.5, and extreme uncertainty for the 2080s under RCP 8.5. These uncertainties notwithstanding, overall, there was a pattern of consistent decline in projected abundance, 2. For the summer period being different across the NNRs (Fig. 8), and declining consistently from the 2040s (Ariundle, Glen Nant and Glencoe), remaining stable and declining from the 2080s (Beinn Eighe), or showing a stepwise decline with overlapping values (Glasdrum), while 3. For the winter season projected abundance declined consistently relative to the baseline across all NNRs. 4. The consistent, non-overlapping decline in projected abundance for the winter season, compared to the summer season (Fig. 8), reflected differences in the statistical distribution of microclimatic conditions. For example, standard deviation of VPD_{month_mean} at the baseline was for Ariundle NNR ± 246.2 or ± 43.6, for Beinn Eighe NNR ± 150.9 or ± 36.2, for Glasdrum NNR ± 230.6 or ± 47.1, for Glen Nant NNR ± 212.1 or ± 44.8 and for Glencoe NNR ± 207.3 or ± 40.4, for summer and winter respectively. Finally, when measuring the proportion of microhabitats within each NNR that had microclimatic refugia within a given distance, 5. There were no projected refugia for the winter season across all NNRs, for any climate change scenario. For the summer season (Fig. 9), 6. Approximately 92 % and 78 % of microhabitats at Beinn Eighe NNR and Glasdrum NNR achieved refugia within 50 m distance for the 2040s (RCP 4.5) scenario, with other sites appearing less resilient, and having <20 % microhabitats achieving refugia over a 100 m distance. 7. While Beinn Eighe NNR remained resilient through to the 2060s (RCP 6) scenario, and Glasdrum NNR appeared less resilient (equivalent values at 50 m distance were 89 % and 38 %, respectively), for the 2080s (RCP 8.5) scenario, the resilience of Beinn Eighe NNR had declined (14 %), while Glasdrum NNR maintained a higher degree of resilience (35 %). The overall pattern was indicative of site-specific levels of microclimatic resilience under progressively more severe climate change scenarios.

4. Discussion

4.1. Temperate rainforest epiphytes and microclimates

Temperate rainforest is a globally rare ecosystem, characterised by its association with a climate that is consistently moist with seasonally cool summers and mild winters (Alaback, 1991; DellaSala, 2011). This climate association implies – as a corollary – that temperate rainforest could be threatened by climate change, and recent studies have pointed to the risks posed by summer warming and drying (Baird et al., 2023; Silver et al., 2024) pushing the current extent of temperate rainforest outside of its climate envelope.

Notwithstanding the climate change effect on foundational components of temperate rainforest – tree composition and stand structure – the unique biodiversity elements that characterise temperate rainforest include an assemblage of lichen epiphytes that are highly restricted to its particular climate (Ellis, 2016; Goward and Spribille, 2005). Although taxonomically different for disjunct temperate rainforests across the planet, many of the species in these assemblages have in common a convergent adaptation to high and frequent moisture (Green and Lange, 1991; Ormond et al., 2025). It is the biomass and diversity of these lichen species – given the link between their adaptative traits and climatic niche – that makes temperate rainforests ecologically distinct. However, lichens are relatively small (typically measured in mm or cm) and therefore strongly nested within forest and woodland microclimates (Coxson and Coyle, 2003; Esseen and Renhorn, 1998; Nilsson et al., 2022). Given the expectation that microclimate specialists such as

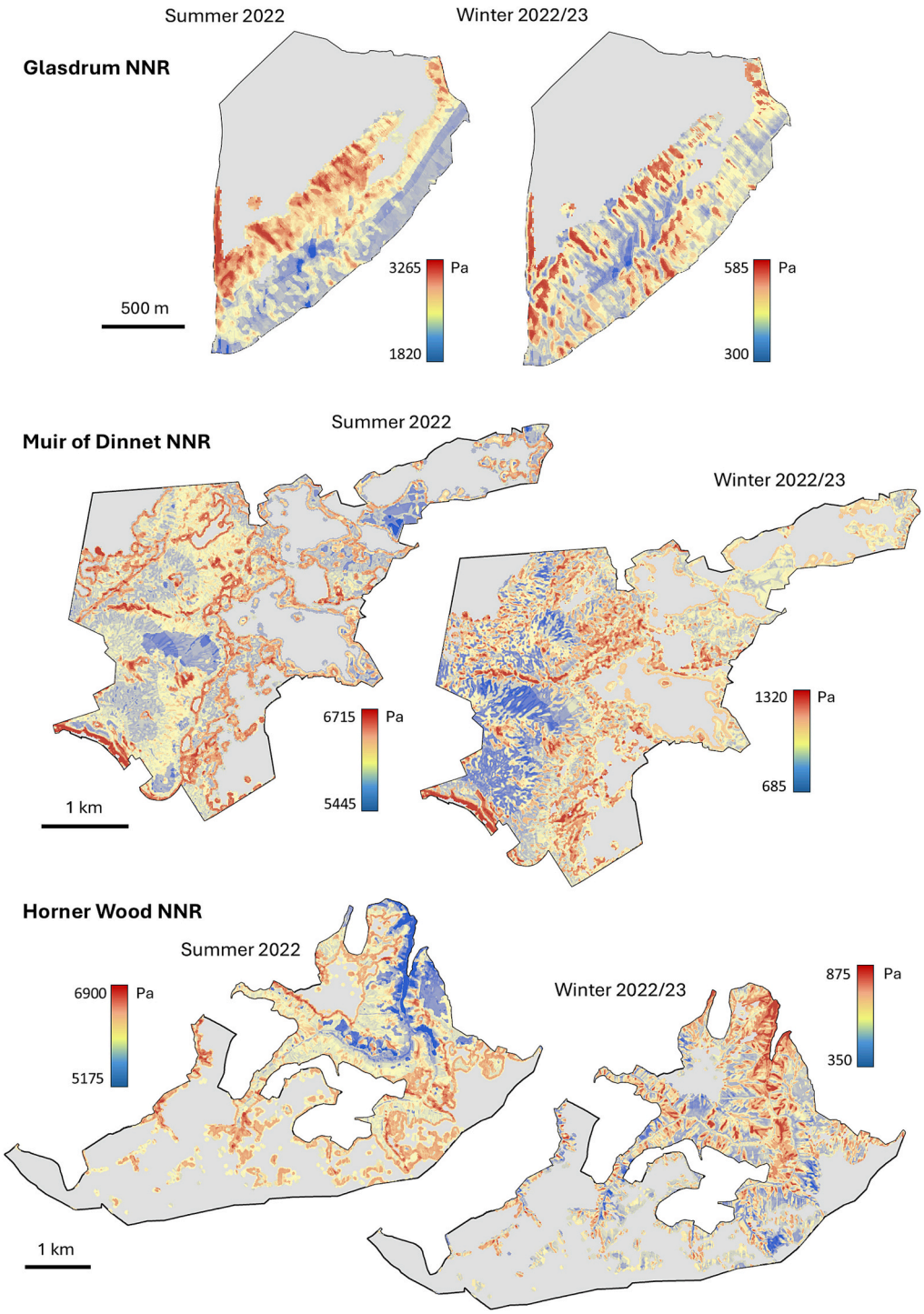


Fig. 4. Projection of random forest models predicting microclimatic VPD_{month_mean} (5 m grid-scale) over woodland areas of the three study sites (grey = no canopy cover), for the datalogging periods of summer 2022 and winter 2022/2023. Note the different scales for each study site and season, necessary to demonstrate spatial contrasts. In colour online.

lichens may be buffered from macroclimatic change within forest and woodland microhabitats (De Frenne et al., 2019; De Lombaerde et al., 2022), microclimatic modelling becomes essential to understand the climate change risk to temperate rainforest biodiversity. Accordingly, previous studies have strongly suggested that the lichen epiphyte response to climate change may depend on landscape topography and stand structure, which – as proxy microclimatic controls – affect thallus growth, abundance of lichen guilds and community scale patterns (Ellis, 2020; Ellis and Eaton, 2021a, 2021b). It was the aim of this study to move beyond proxy habitat features to directly measure and model microclimatic vapour pressure deficit (VPD), a key driver of lichen hydration and physiology (Borge and Ellis, 2024; Stanton et al., 2022). This moisture focus has wider application, since VPD is relevant to a spectrum of ecological processes ranging from stand-replacing fire dynamics (Sedano and Randerson, 2014), through plant hydrological and ecosystem functions (Grossiord et al., 2020; Novick et al., 2024), to microhabitat use by invertebrates (Kaspari, 1993).

Consistent with previous measurement studies (Rambo and North,

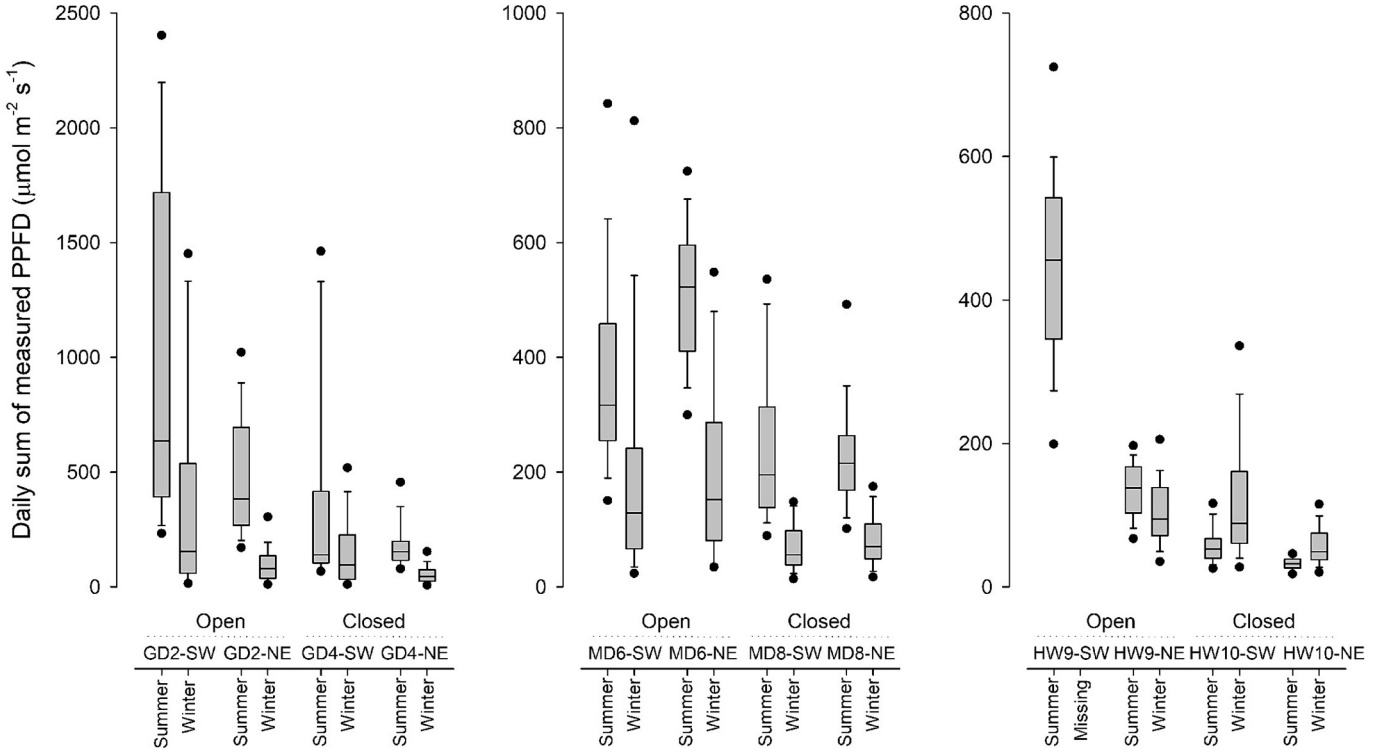


Fig. 5. Boxplots showing the daily sum of PPFD for individual microhabitats with a relatively open and closed canopy for each study site and compared across the summer and winter seasons. Boxplots show the median (line), 25th and 75th percentiles (box), 10th and 90th percentiles (whiskers) and 5th and 95th outliers (symbols).

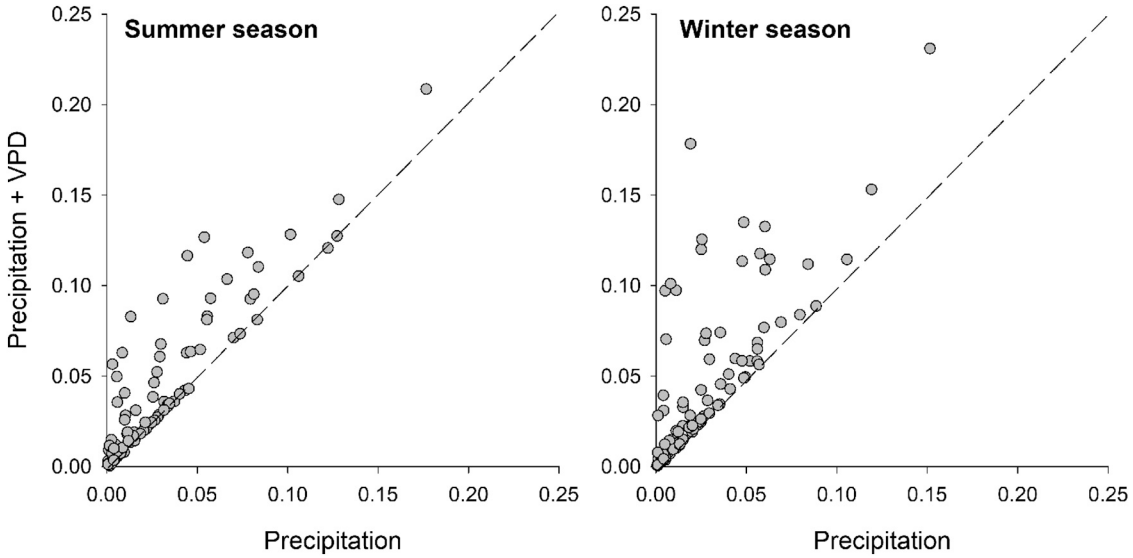


Fig. 6. Comparison of NPMR models for lichen epiphyte species (Table S2) using precipitation alone, or precipitation plus microclimatic VPD_{month_mean} as predictors for abundance (measured across multiple quadrats per tree). Axis values show the cross-validated R^2 for equivalent models, with the dashed line being equity.

2008, 2009), microclimatic VPD was found to be highly variable at microhabitat scales, reflecting a deployment of dataloggers across heterogeneous landscape topography and stand structures. The variance in VPD among stands within a site (measured in 10^1–10^2 m), or even among microhabitats within a stand (measured in 10^1–10^2 cm), at times exceeded variance among sites that were 10^1–10^2 km apart. When summarised as cumulative daily VPD totals (VPD_{day_total}), and furthermore as their monthly means (VPD_{month_mean}), the signature of a predictable diurnal pattern in VPD that will affect cycles of lichen hydration (Borge and Ellis, 2024) appeared to be retained. This facilitates a robust comparison between microclimatic VPD, lichen hydration, and the seasonal patterns of diurnal light availability that drive both photosynthesis (Palmqvist et al., 2010; Palmqvist and Sundberg, 2000) and nighttime metabolic recovery (Alam et al., 2015; Bidussi et al., 2013) at a temporal scale (per month) that can be resolved by lichen growth studies (Borge and Ellis, 2024; Ellis et al., 2017). Although light is not considered here in physiological detail, the use of precipitation and VPD to model the lichen niche (see *4.2 Microclimatic Modelling of VPD and the*

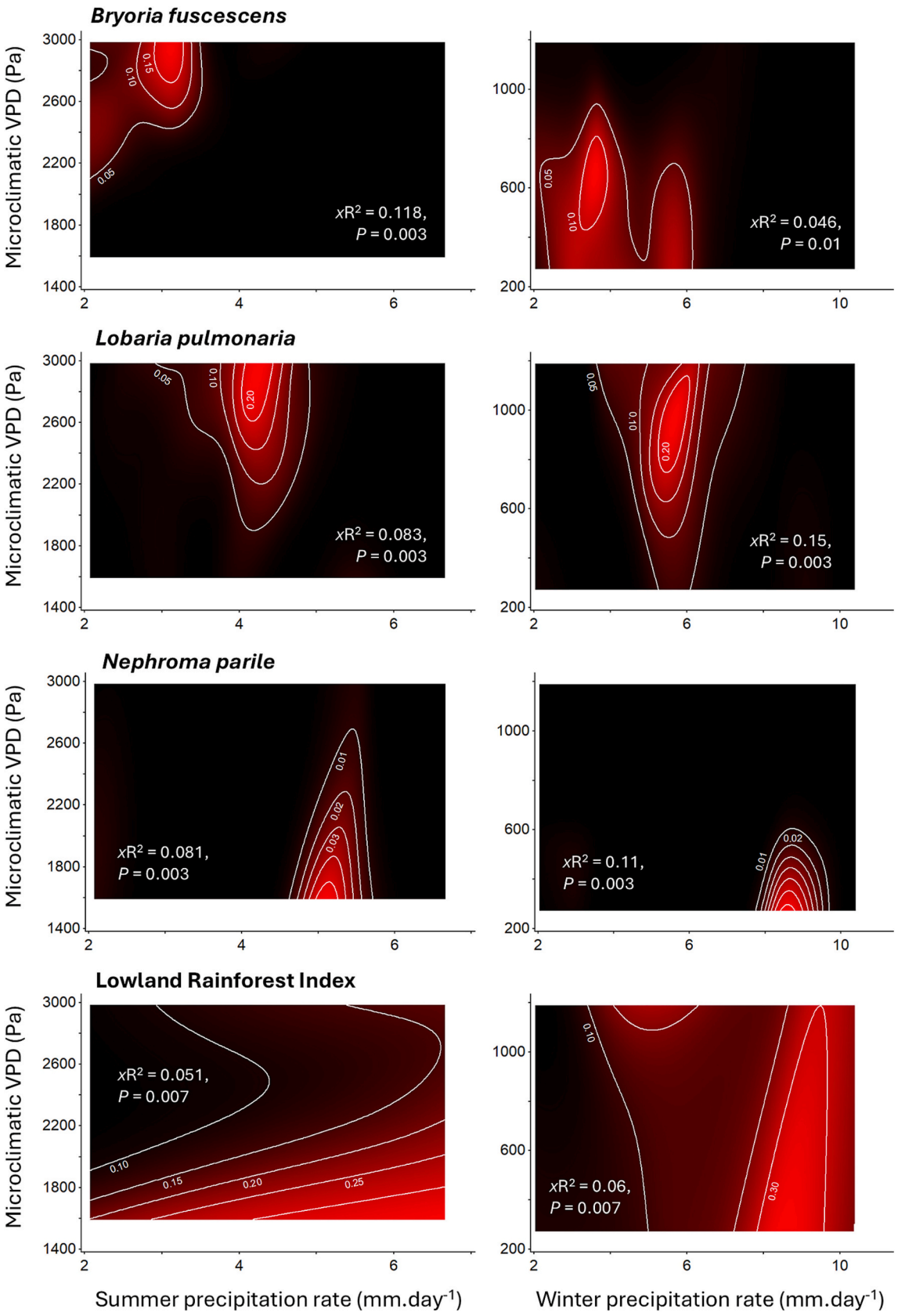


Fig. 7. Contour response plots for NPMR models characterising a relationship between the abundance (measured across multiple quadrats per tree) of ecologically contrasting lichen epiphyte species, and Lowland Rainforest Index indicators, and gradients in summer and winter precipitation rate and their respective values of microclimatic VPD_{month_mean}. In colour online.

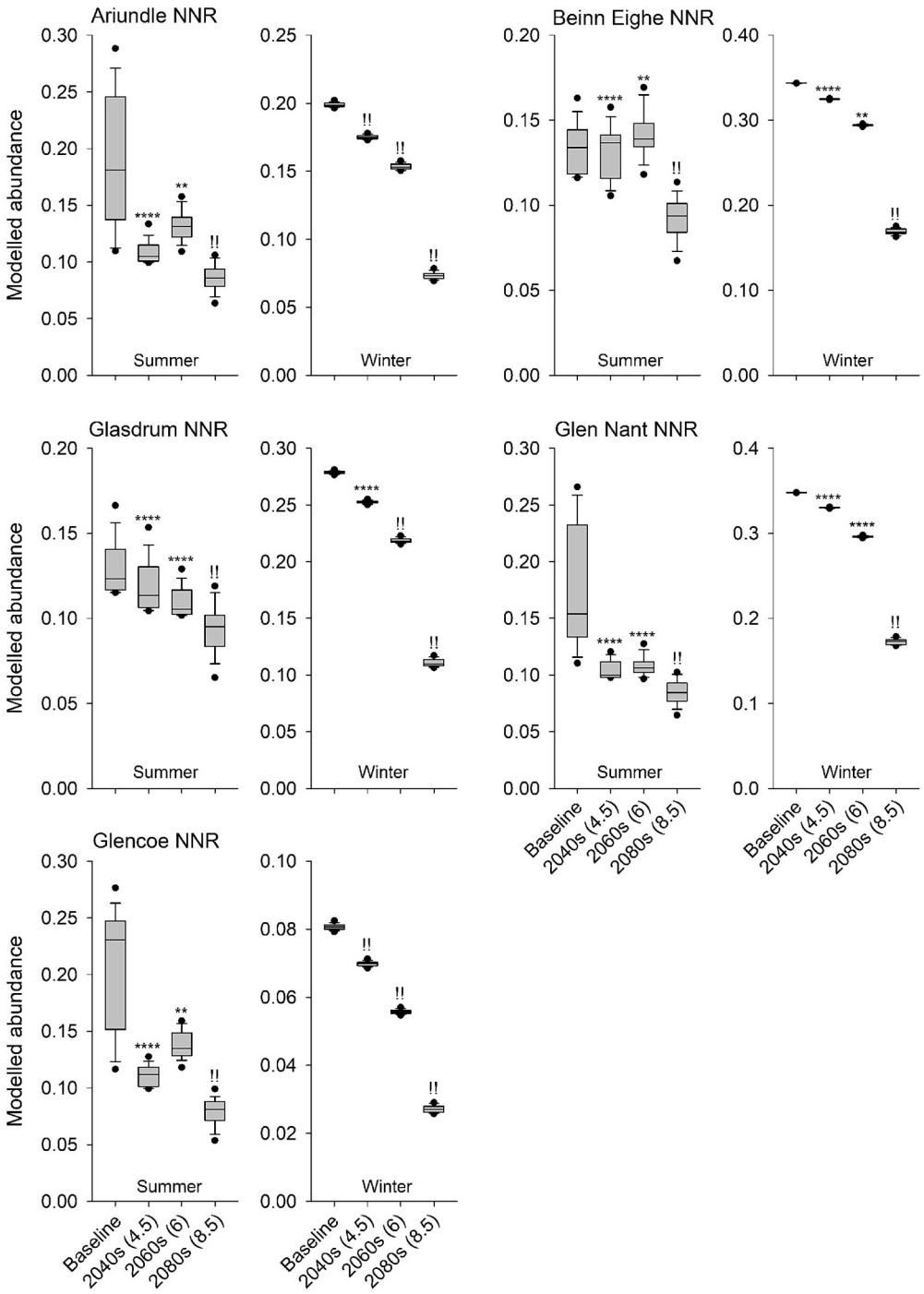


Fig. 8. Projection of NPMR models for a selection of five temperate rainforest NNRs (Fig. 1, inset), showing modelled abundance (measured across multiple quadrats per tree) of Lowland Rainforest Index indicators (Fig. 7) at the baseline and for three increasingly severe climate change scenarios for summer and winter seasons. Boxplots show the median (line), 25th and 75th percentiles (box), 10th and 90th percentiles (whiskers) and 5th and 95th outliers (symbols). Based on the degree of extrapolation into unsampled climate space, projections are indicated as having very high confidence****, high confidence***, medium confidence**, low confidence*, high uncertainty[!], or extreme uncertainty[!!].

Epiphyte Response, below) was applied separately to the summer and winter seasons, which – subject to microhabitat effects – were shown to have higher and lower light levels, respectively. Higher summer and lower winter light – appearing counter to the pattern of closed (summer) and open (winter) canopy – may be an outcome of the relatively open canopy structures of the temperate broadleaf woodland study sites, and their relatively high latitude. Although consistent with previous studies at high latitudes (Larsson et al., 2012) this pattern may contrast with lower latitude sites (Gaio-Oliveira et al., 2004). These seasonal differences in diurnal light regime can strongly affect outcomes for lichens under climate change (Ellis, 2019b), underscoring the potential importance of modelling lichen abundance with due regard to seasonal light regimes.

4.2. Microclimatic modelling of VPD and the epiphyte response

As with comparable modelling studies (Burton et al., 2024; Wang et al., 2025), the statistical downscaling of VPD was founded on

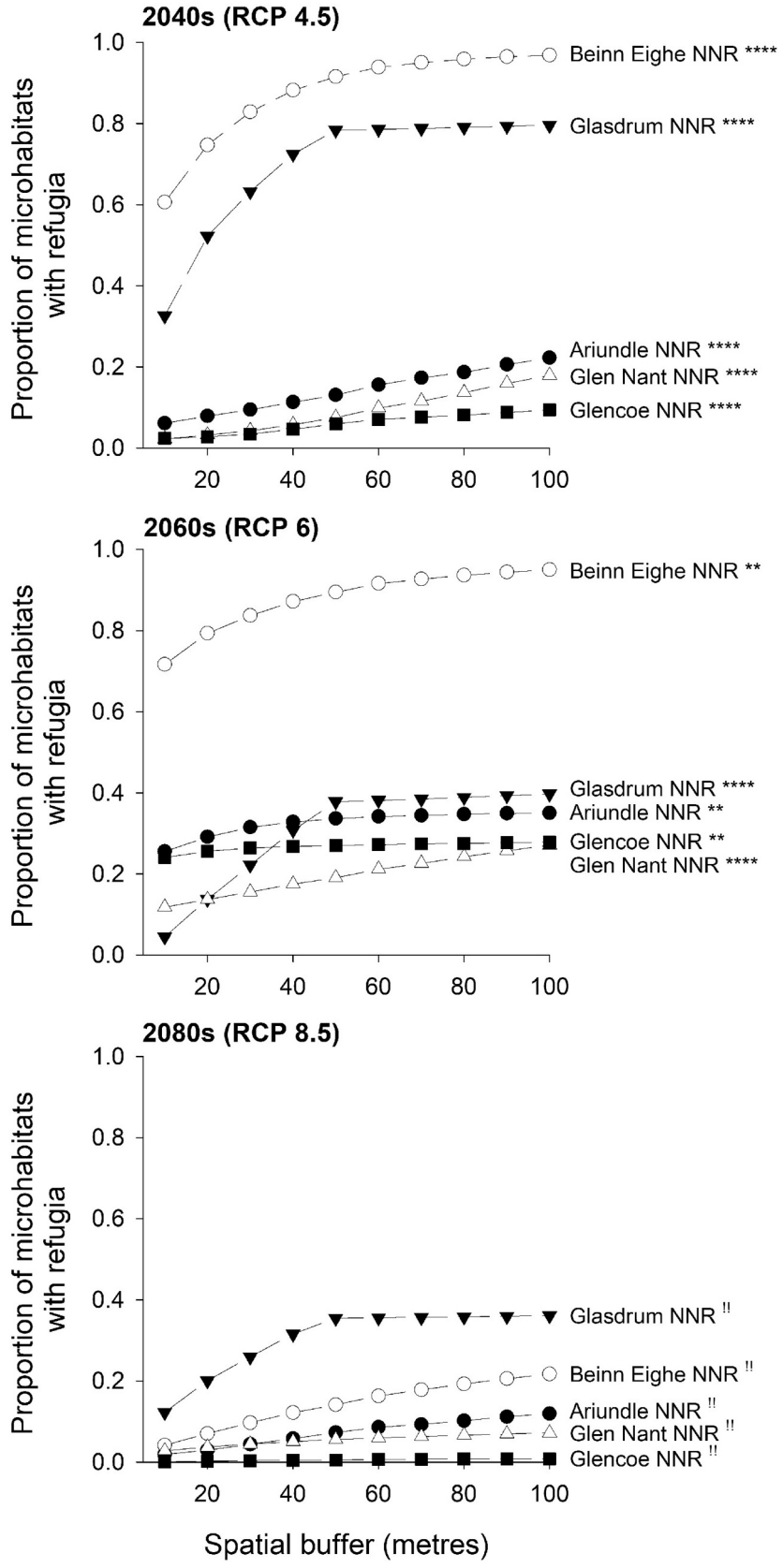


Fig. 9. Graphs based on the projection of NPMR models for Lowland Rainforest Index indicators during the summer season (Fig. 8), used to calculate the proportion of microhabitats that have microclimatic refugia within a given distance (10–100 m), for three increasingly severe climate change scenarios. Based on the degree of extrapolation into unsampled climate space, projections are indicated as having very high confidence****, high confidence***, medium confidence**, low confidence*, high uncertainty[!], or extreme uncertainty[!!].

macroclimatic contrasts spatially and temporally, with precipitation rates driving recharge of moisture into the study sites, affecting humidity, and air temperature controlling the saturated vapour pressure. These macroclimatic effects were then modified locally by altitude (decreasing adiabatic temperature), the redistribution of soil water and heating affected by slope and aspect, and canopy cover (Burton et al., 2024; Davis et al., 2019; Wang et al., 2025). Topographic wetness was consistently more important in explaining VPD than the heatload index, which matches with the known effect of soil water on humidity, plus its additional effect in regulating microclimatic temperature (Davis et al., 2019; Greiser et al., 2024). It was unexpected that during model optimisation (see *2.2 Microclimatic Modelling*, above) microhabitat height and aspect on the tree bole would be dropped as uninformative predictors, considering the known sensitivity of lichens to these variables (Merinero et al., 2015; Rubio-Salcedo et al., 2015). However, the dataloggers measured free air temperature and relative humidity c. 2 cm from the bark surface (Fig. S2), and there may be a stronger effect on epiphytes of direct surface heating, not measured here.

The projected VPD was used to parametrise niche models. Previously, precipitation has been identified as an important variable in lichen bioclimatic models (Bolliger et al., 2007; Cameron, 2021; Nascimbene et al., 2016) and was also adopted here. However, including microclimatic VPD as an additional moisture variable was found to significantly improve model performance, despite over-fitting controls, across a wide range of lichen epiphytes. This is consistent with VPD being functionally related to the likelihood of dew formation, the availability of water vapour, and the drying effect of the air (Anderson, 1936; Barry and Blanken, 2016; Geiger et al., 2012), therefore integrating across different lichen moisture sources (Gauslaa, 2014) with direct consequences for lichen physiological activity and thallus growth. To strengthen these conclusions (see Fig. 7), the niche models based on precipitation and microclimatic VPD appeared ecologically sensible for species with different phenotypes that represent contrasting adaptations to moisture (Gauslaa and Coxson, 2011; Trobajo et al., 2022; Wan and Ellis, 2020). The hair-lichen *Bryoria fuscescens* has a relatively low water holding capacity (mg H_2O cm^{-2}) and will achieve rehydration to optimise photosynthesis under conditions of lower water availability (Solhaug et al., 2021). As a chlorolichen, it has ability to become photosynthetically active under high water vapour content (Lange et al., 1988; Lange et al., 1986) but will be susceptible to supra-saturation (Lange et al., 2000, 2004; Lange et al., 1996) that could ultimately cause population die-back (Gauslaa, 2023). It was therefore positioned by the niche model into low precipitation, more rapidly drying microhabitats. In contrast, the foliose cyanolichen *Nephroma parile* requires liquid water to optimise photosynthesis (Lange et al., 1993) and has a higher water storage capacity (higher WHC) with a greater demand for water to optimise photosynthesis, though with a lower susceptibility to supra-saturation under high moisture conditions. It was therefore positioned by the niche model into high precipitation, more slowly drying microhabitats. The foliose, tripartite *Lobaria pulmonaria* occupies an intermediate moisture strategy (Hovind et al., 2020) and was positioned accordingly.

4.3. Microclimatic refugia and climate change scenarios

Niche modelling was extended to an assemblage of lichen epiphyte indicators that define conservation status for UK temperate rainforests (Sanderson et al., 2018), projecting these models under baseline and climate change scenarios relevant to western Scotland, for five representative NNRs. Variability in outcome for these indicator species, compared among sites, is related to microclimates mediated by site-specific landscape topography and canopy cover. Overall, it was shown that the threat of climate change remained significant despite potential for buffering at microhabitat scales (De Frenne et al., 2019; De Lombaerde et al., 2022). A key feature of the projections was the greater degree of microclimatic buffering and higher resilience of indicator species abundance during the summer compared to the winter season. This is consistent with a seasonally stronger offsetting effect of a closed canopy in maintaining low summer VPD, relative to an equivalent canopy when open during the winter (von Arx et al., 2013; Zhang et al., 2021). Future winter microclimates with excess moisture may render lichens susceptible to supra-saturation and photosynthetic depression (Lange et al., 2000, 2004; Lange et al., 1996), plus sustained and elevated respiration under low winter light but higher winter temperatures (Sundberg et al., 1999; Sundberg et al., 1997). Exploring buffered summer season microclimates in greater detail, lichen epiphyte resilience will depend on the ability of species to disperse locally to colonise suitable microclimatic refugia under climate change. The dispersal of niche specialist lichens is expected to be spatially limited at the decadal timescales relevant to climate change scenarios (Fedrowitz et al., 2012; Johansson et al., 2012). Detailed monitoring studies have shown observed colonisation events limited to average distances of c. 35–97 m over a 9–10 yr interval for *Lobaria pulmonaria* (Belinchón et al., 2017; Öckinger et al., 2005), though being as low as c. 4 m over 37 yr for *Usnea* (*Dolichousnea*) *longissima* (Esseen et al., 2023). The availability of refugia at distances of up to 100 m was estimated, and for three of the five NNRs (Ariundle, Glen Nant and Glencoe) < 20 % of existing microhabitats had future microclimates (2040s, RCP 4.5) that met or exceeded baseline values for species modelled abundance, with this dropping for increasingly severe climate change scenarios to <10 % of microhabitats (2080s, RCP 8.5). These NNRs were ranked fifth (Ariundle NNR – c. 8 ha), fourth (Glencoe NNR – c. 13 ha) and second (Glen Nant NNR – c. 55 ha) for the extent of woodland habitat they contained. Beinn Eighe and Glasdrum NNRs appeared more resilient, with c. 70–80 % of existing microhabitats having future microclimates (2040s, RCP 4.5) that met or exceeded baseline values for species modelled abundance, this being maintained for Beinn Eighe (2060s, RCP 6), then reducing under the most severe climate change scenario (2080s, RCP 8.5) to <20 % of microhabitats (Beinn Eighe) and c. 35 % of microhabitats (Glasdrum). Although having a limited extent of woodland habitat (c. 45 ha), Beinn Eighe appeared to benefit from a more northerly and cooler baseline climate, offsetting climate warming and drying through to the 2060s, while ultimately the larger extent of woodland habitat at Glasdrum (c. 86 ha) appeared to provide additional resilience over the longer-term. The extent of woodland habitat was unrelated to the footprint of each NNR, or their topographic heterogeneity. Glasdrum NNR is c. 168 ha ($\approx$ 51 % woodland cover) and occurs on a predominantly south-east facing hillside, while Beinn Eighe is c. 4857 ha ($\approx$ 0.9 % woodland cover), and Glencoe is c. 5623 ha ($\approx$ 0.2 % woodland cover), both occupying complex mountainous topography and with potential for reforestation targeted towards temperate rainforest refugia that could be informed by patterns in microclimatic VPD and individual species or indicator abundance models.

4.4. Caveats and future areas of work

The results presented are subject to a series of caveats. First, the study is constrained by methodological approaches, from the type of dataloggers used and approaches to their deployment (Maclean et al., 2021), to the statistical downscaling of microclimates. Although positioned beneath the woodland canopy within open hoods (Fig. S2), the dataloggers may have been occasionally exposed to direct sunlight especially where they occurred in edge or gladed settings. Moreover, the statistical downscaling approach has the advantage of being computationally tractable at fine resolutions across the expanse of NNRs, though assumes relationships derived from the model training data, among predictors, and between predictors and response, remain valid when projected to new sites and/or time-periods. This is partly addressed by projecting VPD under climate change scenarios for a suite of northern NNRs (Fig. 1, inset), which are expected to become more similar under their future climates to a southern site (Horner Wood NNR) used for sampling the training data. However, projections for increasingly severe

climate change scenarios demanded model extrapolation that reduced confidence and generated uncertainty. These limits to model transferability are likely further exacerbated if considering additionally challenging moisture variables, such as patterns of orographic or radiative fog (Egli et al., 2019), and more complex moisture distributions, such as across the three-dimensional vertical structure of forest canopies (Rambo and North, 2008, 2009). Addressing such uncertainties may benefit from application of process-based mechanistic models (Maclean and Klinges, 2021; Maclean et al., 2019).

Second, the study assumed that forest structure will remain stationary under future climate change, thereby making it possible to isolate the effect of alternate climate change scenarios on the availability of microclimatic refugia. However, climate change consequences for lichen epiphytes will include any shifts in forest and woodland composition and structure that act as indirect drivers of their microclimate. Plausible but divergent outcomes could include an increase in the extent of forest and woodland canopy cover (Roebroek et al., 2025), through to a loss of canopy cover caused by stressors such as drought (Allen et al., 2010; Senf et al., 2020), with contrasting implications for epiphyte microclimates. An additional important factor not considered here is tree species composition, which is critical for determining the occurrence of lichen epiphytes (Jüriado et al., 2009; Jüriado et al., 2003) and which could be built into projections of habitat suitability (Ellis, 2018; Ellis and Eaton, 2021b).

Third, there is increasing evidence to suggest that lichen epiphytes are not just passively responding to the microclimate but play a functional role in shaping forest and woodland microclimates through the uptake and release of moisture, modifying the water cycle and contributing to evaporative cooling (Pypker et al., 2006a, 2006b; Pypker et al., 2016). There is therefore a relationship between the effect of changing microclimates on epiphyte community composition (also extending beyond lichens to include bryophytes), and the effect of changing epiphyte community composition on forest and woodland microclimates. Consistent with a growing appreciation of the ecosystem roles played by cryptogamic organisms (Porada et al., 2023), future models may benefit from feedbacks between climate change, epiphyte community structure and microclimates.

5. Conclusions

The study has demonstrated the importance of microclimates focussed on the VPD response of lichen epiphytes and therefore measuring the resilience of temperate rainforest biodiversity to climate change. The same methods could be applied to optimise management of woodland structure, and inform patterns of reforestation, for securing future microclimatic refugia. Nevertheless, there are important limitations that remain to be addressed. These include: 1. Dealing with the complexity of lichen moisture sources (including fog and dew) and three-dimensional woodland structures that may benefit from process-based models, 2. Extending the analysis to indirect drivers of forest and woodland microclimates, including shifted stand structure as the trees respond to climate change, and 3. Incorporating epiphytes into the parameterisation of microclimates, considering that they are not simply passive but have an active role to play in moisture cycling and evaporative cooling.

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CRediT authorship contribution statement

Christopher J. Ellis: Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing, Writing – original draft.

Declaration of competing interest

I have no competing interests.

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Data availability

The macroclimatic data are made publicly available via the Centre for Environmental Data Analysis (https://www.ceda.ac.uk/) and via the UKCP toolkit (https://ukclimateprojections-ui.metoffice.gov.uk/ui/home); digital terrain models (DTMs) are made available under a UK Ordnance Survey Open Data Licence via Digimap (https://digimap.edina.ac.uk/); dominant leaf type (DLT) and tree cover density (TCD) are made available by the Copernicus data and information policy regulation (EU) No 1159/2013 for full free and open access; the directly measured temperature and relative humidity values are available from the Research Scotland repository (https://hdl.handle.net/20.500.12594/31885).

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